Problem A. A + B = C Problem

Time limit: 1 second
Memory limit: 1024 megabytes

Given three positive integers p_A, p_B, p_C, Bobo challenges you to find out three infinite binary strings A, B, C with period p_A, p_B and p_C respectively satisfying $A \oplus B = C$, or determine it is impossible to do so.

Please refer to the Note section for the formal definition of period and exclusive or.

Input

The first line of the input contains a single integer T ($1 \le T \le 10^4$), denoting the number of test cases. The description of the test cases follows.

The first and the only line of each test case contains three integers p_A, p_B and p_C ($1 \le p_A, p_B, p_C \le 10^6$).

It is guaranteed that the sum of $\max(p_A, p_B, p_C)$ over all test cases does not exceed 10^6.

Output

For each test case, output "NO" (without quotes) in one line if no solution exists. Otherwise, output "YES" (without quotes) in one line. Then, output three binary strings of length p_A, p_B and p_C in three lines, denoting the first p_A, p_B, p_C character(s) of the infinite strings A, B, C respectively.

You can output "YES" and "NO" in any case (for example, strings "yES", "yes", and "Yes" will all be recognized as a positive response).

Example

standard input	standard output
2 2 3 6 2 3 5	YES 01 011 001110 NO

Note

Let $s = s_1 s_2 s_3 \ldots$ and $t = t_1 t_2 t_3 \ldots$ be infinite binary strings.

The period of s is the smallest positive integer k satisfying $s_i = s_{i+k}$ for all $i \ge 1$.

The exclusive or of strings s and t is given by $s \oplus t$ satisfying $(s \oplus t)_i = s_i \oplus t_i$ for all $i \ge 1$.